Two-stage stochastic programming for autologous CAR-T supply chain under batch-failure uncertainty

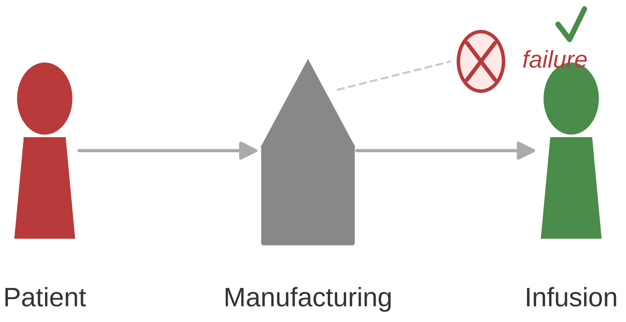
The challenge

4–28%

batch failure rate

(axi-cel, tisa-cel, liso-cel)

UK National CAR-T Panel: 3.87% (Dulobdas 2025)



The approach

Two-stage stochastic program with three-action recourse

Stage 1

plan

Open facilities

Contract capacity

Assign patients

Realize

sample uncertainty

Y, B ~ Bernoulli

200 scenarios

Stage 2

recourse

gated by patient eligibility β_u

The result

5.40% → **2.15%**

−60%

Tier-H cancellation rate

80% → **97%**

+17 pp

Scenarios meeting 95% service target

2.42 M USD

VSS = 14.6%

Mean cost reduction vs. deterministic

Validated on 2,000 out-of-sample scenarios with 90% bootstrap CIs.